

Leakage-Audited Multimodal Federated Learning for Tea Pest and Disease Diagnosis

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Abstract—Tea pest and disease diagnosis draws on both lesion imagery and written field notes, but small agricultural corpora make multimodal and privacy claims easy to overstate. We report a leakage-audited comparison of five vision-language systems on one five-class tea corpus, under source-grouped splitting, exact image-box/text linkage, missing-modality tests, reliability-aware fusion, and measured FedAvg adaptation. The corpus holds 371 crops from 200 photographs, split 222/74/75 across 122/40/38 disjoint sources; every record links to its exact box, and a training-fitted filter masks 178 target-shortcut tokens. On the identical 75-crop support, frozen DistilBERT, frozen ViT-tiny, the raw ResNet-Transformer VLM, a frozen ViT-tiny-DistilBERT VLM, and the routed ResNet-Transformer reach 65.33%, 66.67%, 77.33%, 82.67%, and 81.33% accuracy. Selection is validation-only: the ViT-DistilBERT candidate’s 82.43% validation accuracy stays below the routed incumbent’s 89.19%, so the incumbent is retained. On test the candidate leads by a single crop out of 75, and its higher macro F1 (0.8192 versus 0.6679) traces almost entirely to one crop in a two-sample class. A same-checkpoint diagnostic falls from 92.00% to 33.33% once the note is class-mismatched, and on the locked test deleting the note moves 46 of 75 predictions while replacing the image with noise moves 27, so both branches are used and neither modality is advanced as decisive. The validation-selected ResNet-compact-Transformer VLM retains 98.1–100.0% of its 0.9099 centralized validation macro F1 under warm-start FedAvg over 2–8 simulated clients. Across five federated architectures on 371 genuine box-linked observations, federation outperforms local-only training, although the best aggregator is architecture-dependent; one float32 round costs 18.9–67.2 MiB per client depending on architecture. Two architectures saturate at 1.000 macro F1, which bounds what this corpus can establish; clients are simulated, so multi-site validation remains necessary.

Index Terms—Federated Learning, Vision-Language Models, Tea Pest and Disease Classification, Multimodal Learning, Evaluation Protocol, Data Leakage, Precision Agriculture

I. INTRODUCTION

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five visually confusable stress classes — leaf blight, leafoppers, leaf rust, looper caterpillars, and mosquito bug — reduce yield and leaf quality across major growing regions [1], [2]. Recent diagnostic systems improve field deployment through lightweight CNNs, attention modules, and edge-oriented pipelines [1]–[3], [5], [6], [8], but nearly all begin and end with the photograph. That is not how tea diagnosis is recorded on the ground, where staff note lesion shape, flush condition, weather history, and pest pressure alongside the image [50], [51]. Meanwhile, sending every high-resolution image and field note to a central repository is unattractive for estates with limited connectivity and proprietary records [52], [53]. Federated Learning [15] addresses the second problem, but existing FL work on tea pathology has not integrated the visual and textual evidence used in real inspections.

A second, methodological motivation became dominant during this study. One source photograph may contain several annotated regions, so crop-level random splits place near-identical leaf context on both sides of the boundary. Text can leak the target even more directly when disease names or

class-pure terms remain in descriptions. We therefore group by source image, link text at the exact box level, and learn the blocked vocabulary from training annotations only. This changes the scientific question from “can a model memorise a disease description?” to “does each modality contribute on unseen source leaves?”

Contributions. This is a comparison study, not a systems proposal: no architecture below is advanced as best, and the core carried into the routing and federated experiments is the one validation selected. We provide (i) exact crop-note linkage with source-disjoint splits and train-only leakage control; (ii) common-support unimodal, paired, mismatched, and model-family comparisons, extended by missing-modality and graded-corruption probes that test whether either branch is used at all; (iii) explicit visual and sparse-note routing; and (iv) three-seed client scaling plus a matched evaluation suite covering partitioning, anti-collapse components, warm start, fusion, communication, and four FL aggregators against local-only.

II. RELATED WORK

Image recognition. Tea and plant disease work has moved from backbone benchmarking [3], [4] toward efficiency and deployment: attention-guided lightweight detectors [1], [45], [48], ensembles [6], [7], transformer hybrids [46], [49], real-time field datasets [5], IoT-assisted management [8], edge distillation and quantization [38], [40], and domain adaptation [39], [41]. Surveys map the breadth [2], [9]; across it the work is image-only, centralized, and split at crop or patch level, so one photograph can place near-identical regions on both sides of the boundary.

Agricultural federation. FedAvg [15] is extended with explainability [10], hierarchical sensing [11], blockchain-backed updates [12], cloud coordination [13], and coordinator-free edge federation [14], with trust and continual-update duties reviewed in [42], [47]. Those clients are almost always image-only, and federation is reported as a headline accuracy rather than a measured cost, so communication per round, warm start, realized skew, and matched-budget aggregator comparison are rarely reported together; we report them here.

Vision-language fusion. Contrastive alignment [22], query transformers over frozen encoders [23], gated cross-attention [24], joint captioning [25], and unified interfaces [26] supply the fusion vocabulary, realized here through compact encoders [17], [19], [29]–[33]. Agricultural instances include multimodal LLM fine-tuning [27], federated feature replay over frozen CLIP [28], sensor-image fusion [43], and knowledge-driven semantic classification [44]. Their recurring weakness is the text side: notes generated from the class label or paired by class matching are partial copies of the target. Ours is templated too, which is why pairing is fixed at the exact box index and the shortcut vocabulary is fitted on training text alone. The gap is exactly this: paired image-text FL with a source-disjoint leakage audit.

TABLE I: Source-grouped paired-data audit

Split	Sources	Crops	Blight	Hoppers	Rust	Looper/Mosq.
Train	122	222	77	5	40	61/39
Validation	40	74	26	2	14	20/12
Test	38	75	26	2	13	21/13
Total	200	371	129	9	67	102/64

III. SYSTEM MODEL

Each estate keeps images and notes local; only model updates cross the boundary. The controlled ablations use one selected checkpoint, while FL adapts its trainable multimodal states with the visual backbone frozen. Field-note words are BLAKE2b-hashed into a fixed 30 522-entry ID space and padded or truncated to 128 tokens; ImageNet-normalized 224×224 images become a 7×7 grid of 49 ResNet-50 spatial tokens for bidirectional cross-attention.

A. Dataset and Leakage Audit

Held fixed. This split. Every table in the paper is scored on the same 222/74/75 source-grouped crops, the same exact box–note pairing, and the same train-fitted leakage mask; nothing below re-splits or re-pairs the corpus, so any difference between tables is the axis that table varies and not the data underneath it.

The audited corpus (Table I) contains 200 field photographs collected from the IIT Kharagpur tea garden and annotated with YOLO oriented bounding boxes. One crop is extracted per valid box with 10% context padding, giving 371 examples across *LEAF_BLIGHT* (129), *LEAF_HOPPERS* (9), *LEAF_RUST* (67), *LOOPER_CATERPILLARS* (102), and *MOSQUITO_BUG* (64). Classes are the raw YOLO identifiers under the supplied annotation schema.

Seed 42 splits source filenames, never crops: overlap is zero and the test holds 75 crops from 38 unseen photographs. All 371 records link by exact (photograph, box index); the nine-example leaf-hopper class remains the bottleneck. Notes are templated from manually written phrase pools, not independent farmer prose. A 178-token target-pure vocabulary is learned on training text only and then frozen. No external tea images are pooled; ImageNet supplies weights only.

B. Model Variants and Objective

The reliability network produces (r_t, r_v) , $r_t + r_v = 1$, and the final logits are $\mathbf{z} = \mathbf{z}_f + 2(r_t \mathbf{z}_t + r_v \mathbf{z}_v)$, with a missing modality’s expert zeroed. The residual remains active at test time.

Comparison models. Frozen DistilBERT [33] and ViT-tiny [17] use validation-selected ridge heads. Their late-fusion candidate uses train-fitted 64-component projections and one of 360 validation-screened linear heads; test embeddings are extracted only after the choice is frozen. The raw and routed VLM rows are the cross-attention output before and after the reliability router.

For one-hot label $\mathbf{y}$, training uses class-weighted, label-smoothed cross entropy with auxiliary parent losses and supervised cross-modal alignment:

$$\mathcal{L} = \mathcal{L}_f + 0.20\mathcal{L}_t + 1.25\mathcal{L}_v + 0.05\mathcal{L}_{align}. \quad (1)$$

The visual weight is deliberately larger because image-only learning is the harder path. Modality dropout samples image-only, text-only, and paired batches with probabilities 0.50, 0.10, and 0.40. Checkpoint selection uses $0.65F_{1,\text{paired}} + 0.35F_{1,\text{image}}$ on validation.

For K clients, the same local objective is aggregated with FedAvg [15],

$$\theta^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1}, \quad (2)$$

where n_k is client k ’s sample count. FL starts from the selected checkpoint, freezes ResNet-50, and averages the remaining states after one local epoch; it is adaptation, not training from scratch.

IV. EVALUATED SYSTEMS

A. Base Cross-Modal Checkpoint

The checkpoint trains for at most 15 AdamW epochs (batch 16, peak 10^{-4}). Two cached augmentations per training crop remain views of 222 crops, not new samples. Selection uses 74 validation crops; the strict-reloaded checkpoint is then evaluated once on 75 test crops.

B. Enhanced Image-Only Inference

Held fixed. The encoder. The three pipeline ladders vary the pipeline and hold the model still: the visual ladder fixes the ResNet-50 image path with frozen ViT-tiny as a drop-in rung; the text ladder fixes DistilBERT, the text branch of the system the paper actually retains, so it answers what pipeline engineering would buy on the model being shipped rather than on an arbitrary encoder; and the fusion ladder fixes ViT-Base/16 with DistilBERT, the pair validation selects, ViT-Base leading the five vision encoders at 0.734 validation macro F1 (Fig. 6). Starting from the strongest available pair is what makes a flat result meaningful — a pipeline that cannot help the best pair will not help a worse one. The gate rungs are the one exception: they route among several encoders, which is why they are marked as diagnostics.

Five single deployable visual systems compete on the common support, ranked by the pre-declared validation score $0.6F_1 + 0.4$ accuracy, with the locked test support reported after the choice is frozen. Across the eight visual rungs, locked test accuracy rises from 0.613 to 0.720 (Fig. 5). Figures 2 and 3 show the corresponding text and fusion ladders.

The same ladder is run for text and for text+image on the identical 222/74/75 crop–note support, so every rung is comparable with the others and with the visual ladder. Two differences from the visual case are immediate. At complete notes both ladders are flat — a single-candidate path already reaches 0.987 for text and 1.000 for fusion, while the aligned fusion encoder sweep is likewise saturated (Fig. 7) and the visual ladder gains 0.107 across the same span — so the binding constraint on the note side is the corpus rather than the pipeline, and the validation-to-test gap widens with search (Fig. 4). Figures 2 and 3 show the text and fusion axes. The ladders separate only once notes are degraded, and the ordering is stable — last-four-layer concatenation is the best text rung at both 50% and 75% deletion, deep-feature concatenation the best fusion rung — so representation depth is measurable only after the shortcut is removed. The gates behave as the visual ladder warns: at 50% deletion the three-expert gate has the joint-best validation accuracy (0.932) and the worst test accuracy (0.747).

The calibrated base wins on both sides of the split on the smallest search in the table (validation score 0.771, test accuracy 0.680). The gates score higher only by fitting more routings on 74 crops, and Fig. 4 shows what that costs. We carry the calibrated base as the visual path and report the gate as a diagnostic ceiling.

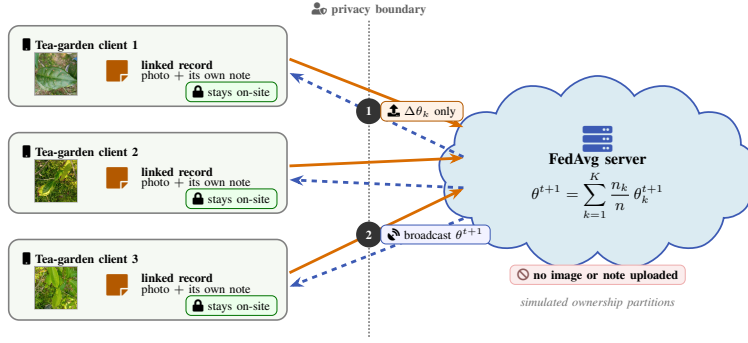


Fig. 1: Federated deployment workflow using representative corpus photographs. Each simulated tea-garden client keeps linked photographs and field notes on-site; only model updates $\Delta\theta_k$ cross the explicit privacy boundary, and the FedAvg server returns θ^{t+1} .

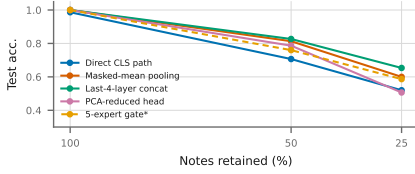


Fig. 2: Text ladder rungs as field notes are deleted, on the identical 75 crops. Every rung falls from the ceiling; gates are dashed and marked *.

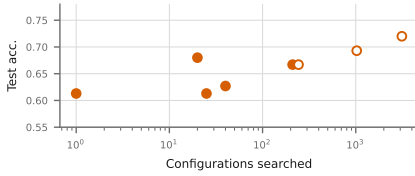


Fig. 5: Visual rungs against configurations searched. Accuracy climbs from 0.613 to 0.720; hollow points are diagnostic gates.

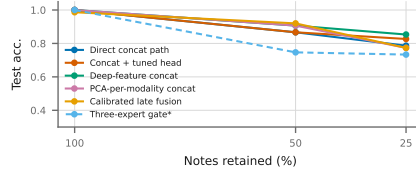


Fig. 3: Text+image rungs under the same deletion. Fusion degrades more gently: the best rung holds 0.853 where the best text rung reaches 0.653.

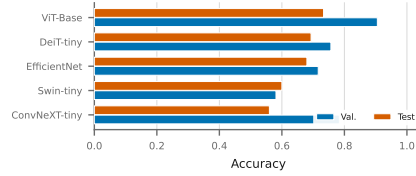


Fig. 6: Image encoders on the aligned support. ViT-Base leads on validation and anchors the fusion ladder.

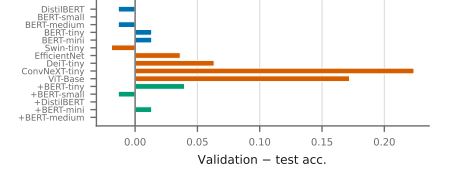


Fig. 4: Validation minus test accuracy per system. A large positive gap is selection on 74 validation crops that did not transfer.

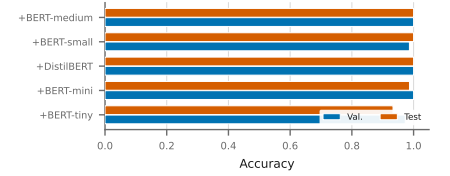


Fig. 7: Fusion encoders on the aligned support with complete notes. All five saturate, so this condition cannot rank them.

C. Reliability-Aware Sparse-Note Rule

For sparse notes, a stable hash retains each non-special token with probability 0.50. A validation-fixed rule routes low-confidence (< 0.60), base-predicted leaf-rust cases to the sparse-text expert; otherwise it keeps late fusion. Because the routing family was refined on this internal partition, its test score is **exploratory and non-pristine**.

D. Multimodal Federated Adaptation

FL adapts 8.69M parameters with frozen ResNet-50, one local epoch, full participation, and sample weighting. We sweep $K = \{2, 3, 5, 8\}$ for eight rounds over seeds 42, 123, and 2026 on the 74-crop validation set. Clients are simulated; the ensemble and router are excluded from this FL score.

The two halves of a round separate cleanly: every stage touching a photograph, box, or note runs on the client.

E. Advisory Module

The Classify–Retrieve–Advise module embeds a query with Sentence-BERT [35] and performs exact FAISS inner-product search [34], [36]. Its separate templated corpus collapses 3,000 annotations to 673 observations; a deduplicated split leaves 472 passages and 201 queries with no shared observation. Correctness requires a class match.

V. PERFORMANCE EVALUATION

A. Protocol

The base checkpoint was trained in PyTorch [37] for at most 15 epochs. All crops from a source photograph remain in one partition. Model selection and the five-expert visual gate use only the 74-crop validation partition; every number below uses the same 75-crop, 38-source internal test support. Accuracy is accompanied by macro F1 because the class distribution is imbalanced. Source-group bootstrap intervals (5,000 replicates) resample source photographs rather than treating multiple crops from one photograph as independent observations.

Five evidence tracks remain separate: the exact-pair same-checkpoint ablation; the validation-selected family comparison on the common 75-crop support; the 74-crop validation-only FedAvg sweep; earlier text, vision, and proxy-fusion screens whose differing supports permit only within-panel ranking; and the 1,000-pair matched-setting evaluation suite. The last uses label-matched resamples and tests optimization and aggregation, not co-observation.

B. Federated Client and Round Scaling

Figure 8 answers the two federated scaling questions directly. Final-round macro F1 ranges from 0.8924 to 0.9099 and is never perfect. At the predeclared $K = 3$ setting, the final mean is 0.9046, or 99.4% of the 0.9099 initialization. Round-wise means stay between 0.8924 and 0.9216; the highest

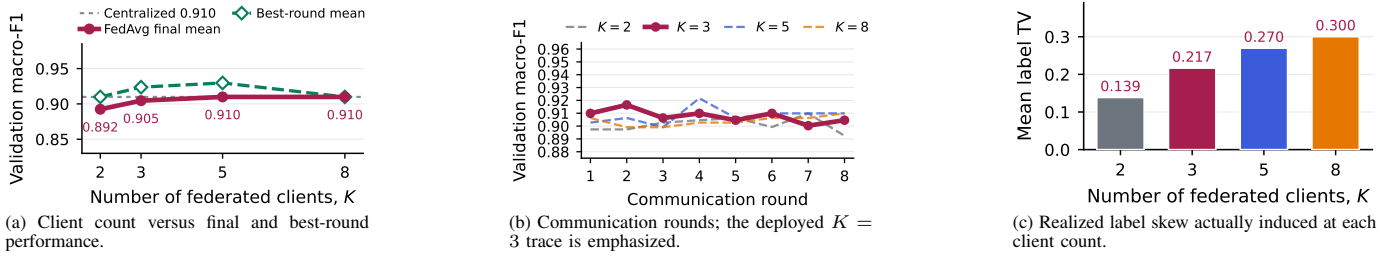


Fig. 8: Measured multimodal FedAvg adaptation on the source-disjoint tea validation split. Points are mean macro F1 over three client-partition seeds; 222 training crops, 74 validation crops, $\alpha = 1$, one local epoch, full participation, and a frozen ResNet-50 are fixed. The dotted line in (a) is the 0.9099 initialization, and (c) is the realized label total variation, so (a) is read against measured rather than assumed skew. The 75-crop test is not used, and client ownership is simulated rather than estate-derived.

transient mean occurs for $K = 5$ at round 4, but it is reported as a validation peak rather than the final model. Increasing K raises realized label-distribution total variation from 0.139 to 0.300 without a monotone final-score penalty.

This stability result is useful but narrow. Every run begins from the same centrally selected checkpoint, so it supports federated adaptation of the multimodal core, not superiority over training from scratch. Three partition seeds measure ownership-split variability, not full checkpoint-training variance, and simulated clients cannot establish estate-level privacy.

C. Matched-Setting Controlled Ablations

Held fixed. The client partition. Within every cell all five architectures train on the identical Dirichlet split, so a gap between rows is architectural rather than a lucky partition; the varied axes are the architecture, the label skew α , and the client count K .

Table II closes the remaining evaluation gaps without mixing supports. The suite runs on 371 genuine box-linked observations with an identical client partition per cell, so differences are architectural. At $\alpha = 0.1$ the corrected splitter realizes label TV 0.451 against 0.265 (Fig. 12), and federation beats local-only training for all five architectures (gains +0.170 to +0.556). Aggregator ranking is mostly a tie; SCAFFOLD is the exception, unstable everywhere (0.088–0.581) and strictly worst in nine of ten cells (Fig. 13). Communication spans $3.5\times$, 18.9 to 67.2 MiB per client-round, so the image-only model reaches 0.520 macro F1 at a third the uplink cost of either fusion model. Four of the five reach exactly 1.000 at $\alpha = 0.5$ and $\alpha = 10$; on a 53-crop validation split that indicates the templated notes are close to label-determining rather than that federation is solved. Warm start does not help once trained out: cold start is better at one seed (1.000 against 0.795) and identical at the other, reversing the eight-round result. The anti-collapse and fusion-variant ablations remain single-architecture.

Federated evidence previously covered one architecture. Table III extends it to the frozen-encoder systems. A closed-form ridge head has no gradients for FedAvg to average, so those systems are federated here as linear probes — frozen encoder, SGD head — which is not the same estimator as the centralized head. Within this protocol the ordering depends on the skew: fusion leads at strong skew (0.677 against 0.571 text and 0.519 image), text at mild skew (0.833 against 0.803). The round-wise crossover occurs at round 33 (Fig. 14). Fusion’s gain over local-only training is nevertheless the largest in both cells (+0.421, +0.372) and grows with clients: at $\alpha = 1$ text leads for $K \leq 5$ but falls to 0.651 by $K = 50$, whereas fusion

holds 0.740 and leads from $K = 10$ up (Fig. 15). Combining the branches helps most where partitioning is most severe. Client partitions remain simulated.

D. Standard Same-Checkpoint Ablation

Fusion is not uniformly worth its cost, and the two blocks say when it is. On complete notes it does not beat its strongest parent — 69/75 against 72/75 for text — so the text path alone is the configuration to deploy, and mismatching the note drops it from 92.00% to 33.33%, confirming cross-modal dependence. Masking half the note tokens reverses the ordering: text alone falls to 64.00% while the router holds 81.33%. The image earns its place only where the note is incomplete, which is the condition field notes are written under. With no note at all, the calibrated visual base gives 68.00%. Two limits apply: notes are templated, so the complete-note block is the most inflated, and the routing family was refined on this partition, with three further hierarchy configurations placing the routed row between 73.33% and 81.33%.

E. Raising Image-Only Accuracy

The validation-selected five-expert gate raises image accuracy from 61.33% (46/75) to **72.00%** (54/75; macro F1 0.6759). This is an ensemble, not the 66.67% pure-ViT baseline; rare-class behavior remains unstable.

F. Unimodal and Multimodal Systems, Centralized and Federated

Held fixed. The method. Every row uses frozen mean-pooled features — no test-time augmentation, no layer concatenation, no calibration, no gating — then one closed-form ridge head whose single penalty is chosen on the 74 validation crops, with plain concatenation for fusion. Four reasons. *Isolation*: under an elaborate pipeline a good score cannot be attributed to the model rather than to the pipeline, so freezing the simplest method leaves the encoder as the only thing that moves. *Determinism*: a closed-form solve has one exact solution, no seed and no initialization, so a gap between two rows is the encoder and not luck, which the TTA and gate rungs discussed above cannot promise. *Minimal search*: one penalty out of six, against the 243–3125 candidates of the ladders; on 74 validation crops that difference is precisely the 0.185 validation-to-test collapse those gates exhibit (0.932 against 0.747). *Lower bound*: plain concatenation is the weakest fusion rung of Fig. 3, so a multimodal gain reported here understates what better fusion achieves, which is why the discussion below points forward to the 0.920 of calibrated late fusion rather than treating the concatenation number as a ceiling. The varied axes are the encoder — classical, then five

TABLE II: Federated results on 371 genuine box-linked crop-note observations (260/53/58 over 153 independent source groups; no synthetic images). Every architecture within a cell trains on the identical client partition, so differences are architectural rather than partition effects. Values are 2-seed means of validation macro F1 over 50 federated rounds. The image-only arm is the only one that never approaches the 1.000 ceiling.

(a) FedAvg by architecture and skew					(b) Aggregator at $\alpha=0.1$					(c) Cost and federation gain		
Architecture	$\alpha=.1$	$\alpha=.5$	$\alpha=1$	$\alpha=10$	Arch.	Avg	Prox	SCAF	BN	Architecture	MiB	Δ local
Image only	0.389	0.557	0.520	0.548	Image	.389	.388	.211	.178	Image only	18.9	+0.170
Text only	1.000	1.000	1.000	1.000	Text	1.000	1.000	.445	1.000	Text only	42.4	+0.421
Concat VLM	0.737	1.000	0.897	1.000	Concat	.737	.772	.253	.742	Concat VLM	59.5	+0.247
Attention VLM	0.988	1.000	1.000	1.000	Attention	.988	.988	.088	.988	Attention VLM	60.5	+0.556
Cross-attn. VLM	1.000	1.000	1.000	1.000	Cross-att	1.000	1.000	.465	1.000	Cross-attn. VLM	67.2	+0.452

TABLE III: Federated adaptation across the paper’s systems, not only ResNet+compact. Frozen encoders with SGD-trained heads on the shared source-grouped split; within a cell every system trains on the identical client partition. Entries are means of 3 seeds over 50 rounds and all client counts, so each is an average of 18 runs; *Local* is the no-communication control. Fusion leads under strong label skew, while the text branch leads at mild skew — the ordering depends on the skew and is reported, not ranked.

System	$\alpha = 0.1$		$\alpha = 1$	
	FedAvg	Local	FedAvg	Local
Text (frozen DistilBERT)	0.571	0.274	0.833	0.466
Image (frozen ViT-tiny)	0.519	0.214	0.639	0.367
ViT+DistilBERT fusion	0.677	0.255	0.803	0.430
ResNet+compact (warm)	0.892–0.910 over 2–8 clients			

text and five vision transformers — and the deployment. Only the federated column departs from the fixed method, because a closed-form solve has no gradients for FedAvg to average and its head must therefore be trained by SGD.

Tables IV and V place a classical feature baseline, five frozen text encoders, five frozen vision encoders, and their fusions on the same 222/74/75 crop-note support, each scored centrally and again under FedAvg. Reading them in that order answers two separate questions: what each modality is worth on its own, and what combining them adds.

Unimodal. With complete notes the text branch is already at the ceiling: frozen DistilBERT reaches 1.000 while the best image encoder, ViT-Base, trails at 0.733. Federating costs little on either side (1.000 to 0.987; 0.733 to 0.667; Fig. 11), so the modality separates these systems, not the deployment mode. The classical rows locate where pretraining earns its place: decisive on images (0.440 for colour histograms with HOG against 0.733), and absent on text, where TF-IDF with a linear SVM is the strongest row at every note sparsity (1.000, 0.933, 0.733). That is direct evidence the note channel is carried by lexical form rather than contextual meaning, consistent with the leakage audit of Section III-A.

Multimodal. At complete notes fusion cannot demonstrate anything: ViT-Base with DistilBERT scores 1.000, the same as text alone, because there is no headroom above a saturated branch. The comparison becomes informative only once notes are incomplete — the realistic field condition. At 75% deletion fusion leads outright, 0.800 against 0.733 for the best image-only encoder and 0.653 for the best text-only encoder, and the same holds encoder-for-encoder at 50% deletion (0.800 against 0.707 and 0.733). Both tables fuse by plain concatenation, the weakest rung of Fig. 3; the deep-feature and calibrated late-fusion rungs reach 0.907 and 0.920 at 50% deletion. Federation preserves the ordering rather than creating it: at 75% deletion federated fusion holds 0.751 while federated text falls to 0.582.

Figures 9 and 10 summarize the encoder-family results under note deletion. Figure 9 shows the classical text baseline

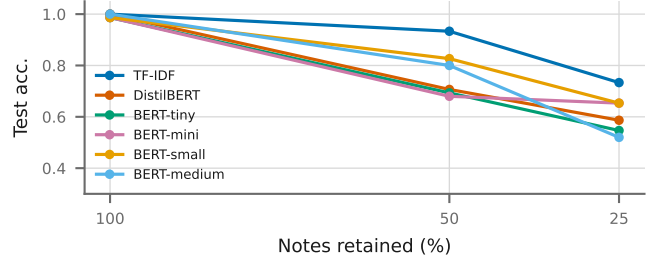


Fig. 9: Text systems as notes are deleted, including the classical baseline: TF-IDF leads every transformer at every sparsity.

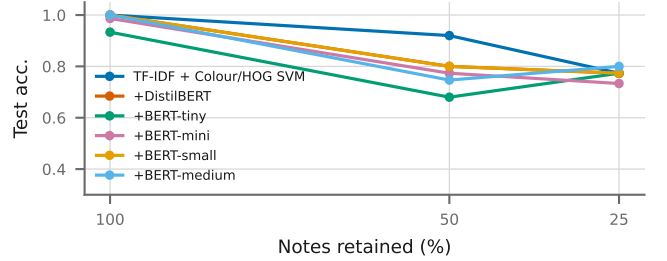


Fig. 10: Fused systems under the same deletion, showing where fusion overtakes both unimodal parents.

leading every transformer at every sparsity; Fig. 10 shows where the fused systems overtake their unimodal parents as notes are removed.

G. Actual Model-Family and Inter-Model Comparison

Held fixed. The support and the selection rule. All five families are screened on the same validation crops under the same pre-declared score, and test is read once afterwards; the varied axis is the model family. The text row alone also fixes the note condition at 50% masking, which is why it must not be read against the complete-note rows of Table IV.

In Table VI the routed ResNet+compact model is the *incumbent* — the core already carried into the routing and federated experiments — and the pooled ViT-DistilBERT model is the *candidate*, adopted only if it beats the incumbent on validation. Validation selects the incumbent before test extraction: 89.19% accuracy and 0.8541 F1 versus 82.43% and 0.6892 for the candidate. The candidate later predicts one more test crop, but the 11-versus-10 discordance is not significant (McNemar $p = 1.0$). The selection/test distinction is what matters here: earlier screens ran on different supports and license within-panel ranking only.

The macro-F1 ordering reverses the accuracy ordering, and one two-crop class produces almost all of it. On leaf hoppers the incumbent scores F1 0.667 on validation and 0.000 on test, while the candidate scores 0.000 on validation and 0.667 on

TABLE IV: Unimodal systems on the common 75-crop test support, centralized and federated. Every row uses the identical 222/74/75 source-grouped split, the identical crop-linked notes, and the identical train-fitted leakage mask; configurations are chosen on the 74 validation crops and test is scored once. Federated rows are FedAvg over $K=5$ Dirichlet($\alpha=1$) clients for 50 rounds, averaged over 3 seeds, with the round chosen on validation only. Centralized heads are closed-form and federated heads are SGD-trained, because a closed-form solve has no gradients to average, so the two columns are close but not the same estimator. Image rows do not move with note sparsity by construction. This table varies the encoder under one fixed pipeline; the ladder study in Section IV-A varies the pipeline instead. Bold marks the best value in each block and column.

System	Dim.	Complete notes				50% of notes deleted				75% of notes deleted			
		Centralized		Federated		Centralized		Federated		Centralized		Federated	
		Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
<i>Text only</i>													
TF-IDF + linear SVM	701	1.000	1.000	0.982	0.925	0.933	0.944	0.911	0.898	0.733	0.683	0.724	0.667
BERT-tiny	128	0.987	0.930	0.889	0.895	0.693	0.673	0.653	0.599	0.547	0.557	0.400	0.372
BERT-mini	256	0.987	0.929	0.982	0.925	0.680	0.668	0.689	0.619	0.653	0.637	0.591	0.519
BERT-small	512	0.987	0.926	0.991	0.973	0.827	0.803	0.680	0.622	0.653	0.628	0.582	0.562
BERT-medium	512	1.000	1.000	0.991	0.953	0.800	0.774	0.658	0.670	0.520	0.396	0.476	0.415
DistilBERT	768	1.000	1.000	0.987	0.949	0.707	0.693	0.618	0.677	0.587	0.528	0.520	0.539
<i>Image only</i>													
Colour+HOG SVM	630	0.440	0.346	0.471	0.392	0.440	0.346	0.471	0.392	0.440	0.346	0.471	0.392
Swin-tiny	768	0.600	0.507	0.569	0.463	0.600	0.507	0.569	0.463	0.600	0.507	0.569	0.463
ConvNeXT-tiny	768	0.560	0.493	0.560	0.470	0.560	0.493	0.560	0.470	0.560	0.493	0.560	0.470
EfficientNet-B0	1280	0.680	0.561	0.680	0.573	0.680	0.561	0.680	0.573	0.680	0.561	0.680	0.573
DeiT-tiny	192	0.693	0.545	0.636	0.520	0.693	0.545	0.636	0.520	0.693	0.545	0.636	0.520
ViT-Base/16	768	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577

TABLE V: Multimodal systems on the same 75 crops, under the same protocol as Table IV. The best text-only and best image-only rows are repeated at the foot for reference. At complete notes the text branch is already saturated, so fusion cannot improve on it and the comparison is uninformative; the separation appears once notes are incomplete, which is the realistic field condition. Bold marks the best value in each column.

System	Dim.	Complete notes				50% of notes deleted				75% of notes deleted			
		Centralized		Federated		Centralized		Federated		Centralized		Federated	
		Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
<i>Multimodal (image + text)</i>													
TF-IDF + Colour/HOG SVM	1331	1.000	1.000	0.987	0.947	0.920	0.939	0.858	0.794	0.773	0.751	0.729	0.673
ViT-Base + BERT-tiny	896	0.933	0.885	0.844	0.779	0.680	0.689	0.756	0.710	0.773	0.634	0.689	0.658
ViT-Base + BERT-mini	1024	0.987	0.989	0.929	0.883	0.773	0.768	0.800	0.769	0.733	0.614	0.729	0.630
ViT-Base + BERT-small	1280	1.000	1.000	0.964	0.971	0.800	0.799	0.742	0.753	0.773	0.633	0.711	0.633
ViT-Base + BERT-medium	1280	1.000	1.000	0.942	0.951	0.747	0.626	0.747	0.727	0.800	0.657	0.702	0.645
ViT-Base + DistilBERT	1536	1.000	1.000	0.920	0.906	0.800	0.793	0.742	0.726	0.773	0.646	0.751	0.752
<i>Best unimodal system, repeated for reference</i>													
Text only: TF-IDF + linear SVM	701	1.000	1.000	0.982	0.925	0.933	0.944	0.911	0.898	0.733	0.683	0.724	0.667
Image only: ViT-Base/16	768	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577

TABLE VI: Common-support encoder selection. Architecture choice uses validation only; test metrics are reported after selection. The text row here is scored under the deterministic 50% sparse-note mask, not on complete notes, which is why it reads 0.6533 while the same encoder reaches 1.000 on complete notes in Table IV; the two agree once the note condition is matched (0.707 there at 50% deletion). The image row uses ViT-tiny rather than the ViT-Base of that table.

Actual system	Val. acc.	Test acc.	Test F1
Text PLM (DistilBERT)	0.6757	0.6533	0.6732
Image ViT (ViT-tiny)	0.7973	0.6667	0.5830
Raw ResNet+compact VLM	0.8378	0.7733	0.7155
ViT+DistilBERT VLM	0.8243	0.8267	0.8192
Routed ResNet+compact	0.8919	0.8133	0.6679

test — the two models swap a single crop, which accounts for 0.133 of the 0.151 test macro-F1 gap. Dropping that class, four-class macro F1 is 0.901 against 0.862 on validation and 0.835 against 0.857 on test, so the lead changes hands by 0.02–0.04. The incumbent’s validation-to-test accuracy movement (66/74 to 61/75) is 1.35 standard errors. Macro-F1 differences on this support are therefore reported and not ranked.

H. Cross-Modal Analysis and Evidence Boundary

Mean reliability is 0.575 text/0.425 image; mismatching text costs 58.67 accuracy points and 0.5949 F1. Exact cross-

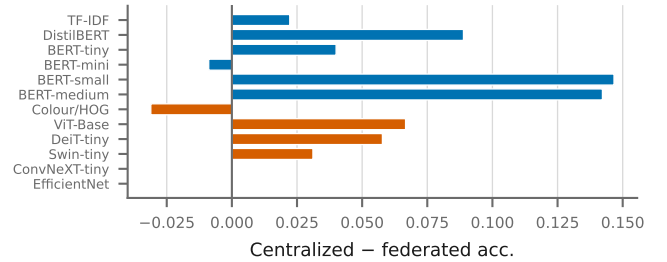


Fig. 11: Centralized minus federated accuracy per encoder at 50% note deletion. Federating is close to free for the image encoders.

modal recall@1 is only 1.33%, while class-level recall@1 is 0.600/0.613, indicating class organization rather than crop identity.

I. Missing-Modality and Corruption Probes

Held fixed. The checkpoint. One trained model is held constant and the *input* is varied instead — each modality corrupted in turn and the image ablated outright — so a change in prediction is attributable to the evidence the model consumes rather than to retraining.

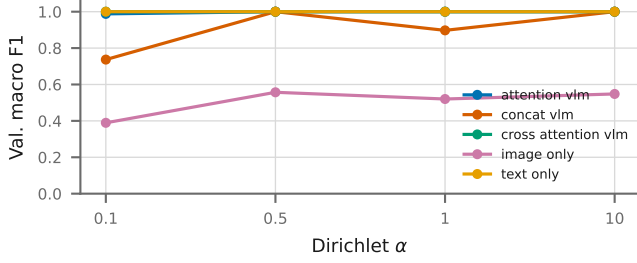


Fig. 12: All five architectures across skews under the corrected class-wise splitter: accuracy falls as α tightens, as intended.

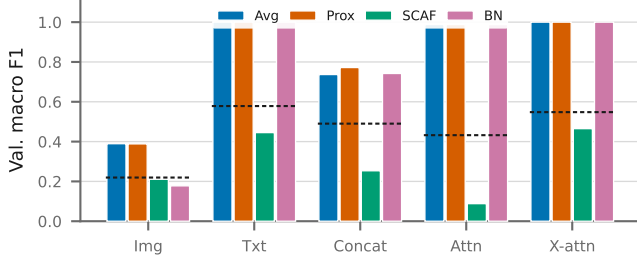


Fig. 13: Aggregators at $\alpha = 0.1$; the dashed rule in each group is that architecture’s local-only control. SCAFFOLD collapses to 0.088.

The mismatch result above shows dependence on the note; it does not show whether the image is used at all. We therefore evaluate the retained checkpoint on the locked 75-crop test under graded corruption of each modality and under two full image ablations, comparing predictions crop-by-crop against the clean run.

Both branches are load-bearing. Table VII reports the sweep. Deleting every attended text token costs 0.551 macro F1 and moves 46 of 75 predictions; replacing the image with uniform noise costs 0.222 and moves 27. The reliability weight is not constant: mean w_t is 0.575 on clean input, 0.481 under complete text deletion, and 0.615 under complete image noise. The note carries more of the decision than the image, but neither branch is inert. No ordering of modalities or architectures is advanced from a single 75-crop support.

Ablation strength determines the verdict. Additive Gaussian noise at $\sigma=0.35$ leaves leaf and lesion plainly visible and moves only 9 of 75 predictions, whereas zeroing the image moves 18 and replacing it with uniform noise moves 27. An earlier 30-pair proxy screen used the Gaussian setting alone and read the visual branch as inert; that reading does not survive genuine removal on the locked support. Evidence that a branch is unused therefore requires an ablation that removes it rather than one that degrades it.

J. Architecture Component Ablation

Nine variants were retrained on the same split in CPU and H100 runs. Although the H100 full model reaches 94.67%, the identical CPU run gives 85.33%; the mean gap is four points and five of eight effects reverse sign. The sweep therefore reports instability, not a component ranking.

K. Advisory Retrieval Results

Retrieval reaches 90.05% correct-class precision@1 against a 20.01% chance rate, rising to 95.52% with the classify stage. That second number is the classifier’s accuracy, because

TABLE VII: Modality corruption on the locked 75-crop test, retained checkpoint. Flips count predictions differing from the clean run.

Condition	Acc.	Macro F1	Flips
Clean	0.9200	0.8753	—
Text deletion 25%	0.8133	0.7445	10/75
Text deletion 50%	0.7333	0.6093	21/75
Text deletion 75%	0.6800	0.6275	24/75
Text deletion 100%	0.3600	0.3240	46/75
Image Gaussian $\sigma=0.09$	0.8533	0.7840	7/75
Image Gaussian $\sigma=0.18$	0.8933	0.8567	9/75
Image Gaussian $\sigma=0.26$	0.8667	0.8279	11/75
Image Gaussian $\sigma=0.35$	0.8933	0.8513	9/75
Image zeroed	0.8267	0.7544	18/75
Image uniform noise	0.6533	0.6535	27/75

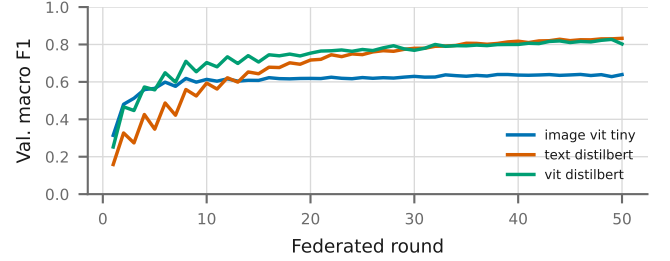


Fig. 14: Per-round convergence of the frozen-encoder systems at $\alpha = 1$, three seeds. Text overtakes fusion at round 33.

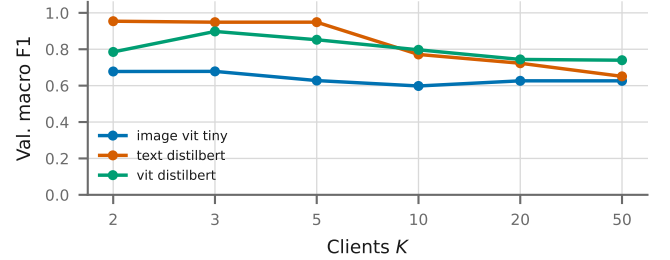


Fig. 15: The same systems against client count. Fusion leads from $K = 10$ upward, where the shards become small.

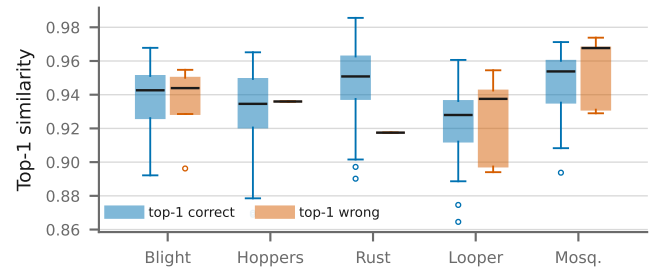


Fig. 16: Top-1 retrieval similarity per class, split by whether that hit was correct. The distributions overlap, so similarity does not separate a correct advisory from a wrong one.

retrieval restricted to one predicted class can only return that class, so routing rather than ranking decides correctness. A stratified query audit locates the failure: open retrieval is near-saturated (99.24%) on the queries carrying a class-unique sentence and falls to 72.46% on the rest, which routing lifts to 89.86%.

Three limits apply. The corpus is template-composed, so base and queries share sentence vocabulary even though no

observation is shared — visible in Fig. 16 as uniformly high similarity — and these are upper bounds relative to free-form notes. The fusion re-ranker, the OBB overlay and the agronomic correctness of the advice remain unevaluated. This is a retrieval result, not evidence that the treatment is right.

L. Reproducibility Manifest

The updated bundle records data identity, checkpoint state, every federated round, and each realized client partition. It separates mechanically verified controls from properties outside the evidence. Deterministic grouping, strict reload, and validation-only adaptation are reproducible; external estate transfer is not inferred from them.

M. Class-Level Error Accounting

The enhanced gate gains eight correct crops and the exploratory router gains 15, mainly in blight and looper; two leaf-hopper examples remain too few for a stable claim.

N. Threats to Validity

The locked test is small (38 sources, 75 crops, two leaf-hopper crops), notes are templated, and selecting five experts on 74 validation crops can overfit. The central checkpoint has one seed; FL seeds vary only client partitions. Clients are simulated, and the matched suite uses class-matched resamples with synthetic image fill rather than co-observations; E2/E3 use one seed. The router result is non-pristine, DistilBERT is a PLM rather than a generative LLM, and earlier fusion screens on proxy pairs have incomparable supports, so their patterns are read without their levels. The corruption probe is one seeded realization per severity on 75 crops and carries no interval. Deployment requires multi-estate data, independent notes, rare-class coverage, and repeated end-to-end training.

VI. CONCLUSION

On 371 linked crops, validation retains the ResNet-compact-Transformer router; the ViT-DistilBERT candidate's one-crop test edge is non-significant ($p = 1.0$), and CPU/H100 component effects are unstable. Warm-start FedAvg retains 98.1–100.0% of initialization over 2–8 simulated clients. In the matched suite, FedAvg averages 0.823/0.884 across the five architectures at $\alpha = 0.1/1$, warm start no longer helps, and updates cost about 60 MiB. No E3 arm collapses, so the stack is not claimed necessary. Multi-estate, co-observed evaluation remains required.

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